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Effect of biochar and organic amendments on improving soil quality and apple orchard productivity: a 2-year field study

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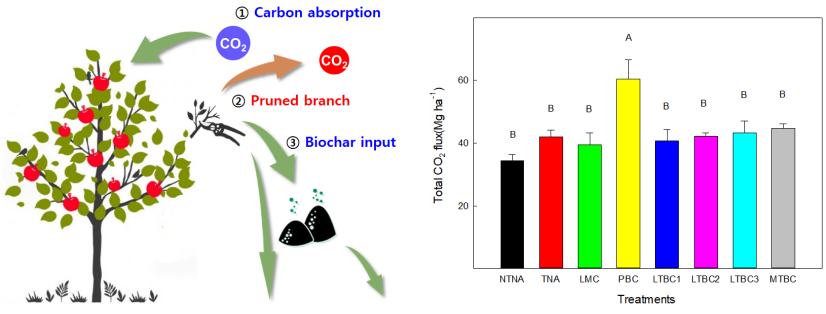
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ABSTRACT

Soil degradation poses a significant challenge to mature apple orchard productivity. This study aims to investigate the effects of organic amendments on soil properties, carbon management, greenhouse gas emissions, apple tree growth and fruit quality in South Korea. We supplied organic amendments, including livestock manure compost (LMC), pruned branch chip (PBC), and three concentrations of low temperature biochar (LTBC; 19.8, 99, 198 m^3 ha^{-1}), as well as mid temperature biochar (MTBC) to the mature apple orchard field over two years. Upon various organic amendments, LTBC significantly improved soil physical and chemical properties, enhancing the capacity for cation exchange and water-holding, and reducing bulk density. Moreover, LTBC demonstrated lower CO_2 emissions compared to PBC, contributing to effective carbon sequestration and GHG mitigation. No significant differences were observed across treatments on nitrous oxide emissions. Interestingly, both LMC and LTBC boosted apple tree growth, development and fruit yield compared to other organic supplements, which may contribute to enhancing fruit nutrient contents. These findings highlight LTBC's ability to improve soil carbon storage and nutrient retention due to its high stability and resistance to decomposition. Identifying key organic amendments from this study will pave a way for reducing dependence on chemical fertilizers while enhancing orchard productivity. This study lays the groundwork for more research into how organic amendments work together and whether they are cost-effective for large-scale use, which will help apple production last and restore the soil.

Keywords: Organic amendments, Biochar, Livestock manure compost, Soil carbon management, Yield



Organic amendment LTBC enhanced soil properties, carbon sequestration, and reduced GHG emissions, while LMC and LTBC boosted apple tree growth and yield.

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Introduction

Nowadays, the production of commercial apples has gained increasing attention in Asian countries like China, India and Japan (Statista, 2023), mainly in South Korea as the leading fruit crop, with an annual production of 422,115 metric tons in 2021 (Geleta et al., 2025). Apples are a globally significant fruit, with higher nutritional value and their cultivation plays a key role in the agricultural economy (Sharma et al., 2025). Soil quality is influenced by factors such as soil depth, organic matter, and electrical conductivity leading to salinization, compaction, and nutrient depletion (Dalal et al., 2011; Lal, 2015). Additionally, soil deterioration, contamination, and fertility loss threaten global food security and agricultural sustainability (Wu et al., 2023). In South Korea, mature apple orchards face several challenges that limit productivity, including soil degradation, nutrient imbalances, and reduced soil carbon content. Furthermore, recent studies on fertilizer use trends in Korean orchards indicate that excessive inorganic fertilizer application has led to nutrient imbalances, reinforcing the necessity of integrating organic amendments for sustainable apple production (Lee et al., 2024a). These issues are exacerbated by conventional agricultural practices, such as the overuse of chemical fertilizers, which can lead to soil compaction, reduced water-holding capacity, carbon storage, and an overall decline in soil health (Lee et al., 2006; Lee and Kim, 2009; Rathinapriya et al., 2025). In particular, soil degradation often affects the growth and yield of apple trees, contributing to lower productivity compared to global standards. Hence, the management of soil carbon plays a vital role in sustaining soil fertility, retaining nutrients, improving water holding capacity, and supporting overall tree health (Singh et al., 2024).

To combat these challenges, farmers are increasingly adopting organic amendments as a sustainable alternative to conventional fertilization methods. Organic fertilizers are effective in soil management practices for enhancing soil fertility and crop productivity (Cheol Kim et al., 2018). The application of organic amendments, such as livestock manure compost (LMC), biochar (BC), and pruned branches, has become widely recognized as part of sustainable agricultural practices. LMC is a nutrient-rich organic amendment that enhances soil structure, nutrient retention, and carbon storage. Its decomposition produces stable carbon forms, reducing soil erosion risks and improving water retention (Goldan et al., 2023). Similarly, incorporating pruned branch chips (PBC) into orchards minimizes waste while enhancing soil carbon storage, structure, and nutrient cycling. These branches decompose slowly, providing a steady nutrient release for trees (Anthony, 2013). BC has recently emerged as a cost-effective and eco-friendly strategy for improving soil quality. BC is a very stable carbon-based substance that is made by breaking down feedstocks, such as agricultural residues, food waste, and forestry waste, using heat (Tomczyk et al., 2020). Its micro-porous structure and high cation exchange capacity improve water retention, nutrient availability, and microbial activity without exacerbating environmental conditions.

These amendments are known for their potential to improve soil physical properties and enhance the soil microbial community (Schulz et al., 2013; Khorram et al., 2015; Khorram et al., 2016). The benefits include increased nutrient levels in the soil, improvements in soil chemical properties, such as cation exchange capacity, and enhanced water and nutrient-holding capacity. Additionally, they help lower bulk density, contributing to better

soil structure (Lehmann et al., 2011a; Schulz et al., 2013; Vaccari et al., 2015; Khorram et al., 2016). Research has shown that the use of BC and compost can significantly improve soil quality, leading to better plant growth and potentially higher fruit yields, predominantly in soils that are poor, acidic, or prone to nutrient leaching (Khorram et al., 2016; Safaei Khorram et al., 2019). These amendments enhance soil fertility and support not only enhance soil fertility but also support more sustainable orchard management by mitigating soil degradation, improving long-term soil health, reducing greenhouse gas emissions, promoting apple productivity, and supporting sustainable agriculture.

Organic amendments have numerous benefits for agriculture but can also contribute to nutrient eutrophication and greenhouse gas emissions, including CO_2, methane, and nitrous oxide (IPCC, 2007). While methane, and nitrous oxide emissions are less abundant, they have a greater climate impact than CO_2 (Rodhe, 1990). Therefore, analyses of various soil amendments facilitate the identification of organic fertilizers that consistently respond to fluctuating environmental conditions.

So, the study's goal is to find out how LMC, PBC, LTBC, and MTBC affect soil carbon management, as well as how they might affect greenhouse gas emissions and apple fruit quality. The end goal is to find the best organic fertilizers for South Korea's mature apple orchards. Perhaps, this study paves the way for future investigation of synergies between different organic amendments and their collective impact on orchard soil health. Additionally, future studies should focus on the economic viability of large-scale applications of these organic materials, especially BC, to determine the most cost-effective approaches for farmers.

Materials and Methods

Experimental site and organic material treatment This study was conducted in an apple orchard located at the National Institute of Horticultural and Herbal Science in Iseo-myeon, Wanju-gun, Jeollabuk-do, South Korea (N 35°.83', 127° 03'). Eight-year-old 'Fuji'/M.26 apple trees were planted at a spacing of 3 m × 4 m. On May 7, 2021, organic amendments were incorporated into the soil. Soil treatments designed as no tillage+no addition (NTNA), tillage+no addition (TNA), livestock manure compost (LMC), pruned branch chips (PBC), low temperature biochar (LTBC), and mid temperature biochar (MTBC) treatments. For the LMC treatment, 20 kg per tree was applied to the soil surface according to conventional practices. Pruned apple branches were shredded, and the following materials were applied: 120 L (26.6 kg) of PBC, 24, 120, and 240 L (2.1, 20.7, and 41.4 kg) of LTBC (pyrolyzed at 350°C), and 120 L (21.0 kg) of MTBC (pyrolyzed at 450°C). PBC amendments were applied in trenches (120 cm wide, 80 cm long, 60 cm deep) excavated 60 cm from both sides of the tree rows and mixed with soil. For comparison, an NTNA control (without organic amendments) and a control (TNA) with soil refilled without organic amendments were included. The physical and chemical properties of the organic amendments were shown in Table 1. LTBC and MTBC were produced by pyrolyzing crushed apple pruning wood at 350°C and 450°C, respectively. LTBC was pyrolyzed in a continuous process for 40 minutes, while MTBC was pyrolyzed in a batch process for 2 hours. The pH, EC, phosphorus, potassium, magnesium, and calcium contents of MTBC were

higher than those of LTBC. This is because the higher pyrolysis temperature of MTBC leads to the concentration of various salts originally present in the raw material (Cantrell et al., 2012; Kim et al., 2022).

Table 1. The physical and chemical properties of various organic amendments used in this study.

Organic amendments	Water content (%)	pH (1:20)	EC (1:20) (dS m^{-1})	P (%)	K (%)	Mg (%)	Ca (%)	Na (%)	TC (%)	TN (%)
LMC	72.2	6.21	25.1	1.08	0.95	0.51	1.63	0.40	38.83	2.58
PBC	22.3	7.03	74.1	0.08	0.46	0.09	0.75	0.01	48.99	0.88
LTBC	4.1	7.16	45.3	0.15	0.85	0.12	0.74	0.05	28.72	1.00
MTBC	4.6	8.24	92.8	0.25	1.13	0.19	0.92	0.03	27.68	1.29

Soil respiration measurement Soil respiration was measured using the closed chamber method. Chambers (40 cm × 25 cm diameter × height) were installed at the sites of organic amendment application. After 7 days of organic amendment treatment, we collected gas samples weekly between 10:00 and 11:00 AM. Gases were collected from the chambers for 30 minutes and were analyzed for CO_2 and N_2O concentrations using gas chromatography (Clarus 680, PerkinElmer, USA). Methane emissions were negligible and therefore not analyzed. A flame ionization detector (FID) with a methanizer was used for CO_2 analysis, and a 63Ni electron capture detector (ECD) was used for N_2O analysis. The gas emission rates were determined by the increased concentration of gases inside the chamber for 30 min and were calculated using the formula (Rolston, 1986)

$$\text{Gas emission rate (mg m}^{-2}\text{ hr}^{-1}) = \rho \times H \times \Delta C / \Delta T \times 273/T \quad (1)$$

where ρ - gas density under the standard state (CO_2-1.22 kg m^{-3}, N_2O-1.98 kg m^{-3}, and CH_4-0.657 kg m^{-3}), H-chamber height (m), $\Delta C/\Delta T$ - gas concentrations difference (mg m^{-3} hr^{-1}) inside the headspace before and after chamber closing, and T - absolute temperature (K) is 273+temperature (°C) inside the chamber.

Singh et al., 1999, the total fluxes of each gas were determined by following formula,

$$\text{Total flux (kg ha}^{-1}) = \Sigma n\, i\, (R_i \times D_i) \quad (2)$$

where R_i - daily flux (mg m^{-2} d^{-1}), D_i - day interval between the i-1st and ith samplings, and n - number of sampling times.

Soil physical and chemical properties After organic amendments were added, the soil's physical properties were checked. These included its resistance to penetration, bulk density, and effective soil depth. Soil physical properties, including soil penetration resistance, bulk density, and effective soil depth, were evaluated after the application of organic amendments. Soil penetration resistance was determined using a digital cone penetrometer

(DIK-5532, Daiki, Japan). Soil chemical properties were analyzed following the methods of the National Institute of Agricultural Science and Technology, RDA, South Korea (NLAST, 2000).. Soil samples were collected from the amendment application sites, air-dried in the shade, and sieved through a 2-mm mesh. Soil pH and electrical conductivity (EC) were assessed at a 1:5 soil-to-distilled water ratio using an ion electrode method (ORION VersaStar Meter, Thermo, USA) for pH and an EC meter (CM-30R, TODKK, Japan) for EC. Organic matter content was determined by the Tyurin method. Ammonium nitrogen and nitrate nitrogen were measured using the Kjeldahl distillation method. Available phosphorus was analyzed using the Lancaster method, and exchangeable cations (K, Ca, Mg) were extracted with a 1N CH_3COONH_4 buffer solution (pH 7.0) and analyzed using inductively coupled plasma spectrometry (Integra, GBC Scientific Equipment, Australia).

Tree growth and fruit productivity assessment Tree growth was assessed in May 2021 (pre-experiment) and March 2023 (post-experiment). Fruit characteristics were analyzed in 2021 and 2022, as their development and harvest followed seasonal cycles within the study period. The vegetative morphological characteristics such as tree height, canopy width, and trunk circumference were determined using measuring tape. Trunk circumference was recorded at 90 cm above ground level. Fruit length and width were measured using a Vernier caliper, while firmness was assessed with a texture analyzer (LS1, AMETEK, USA). Soluble solids content extracted from the fruits was determined using a refractometer (PAL-1, ATAGO, Japan). The fruit acidity was determined using an automatic titrator (Titroline 5000, SI Analytics, Germany) with 0.1N NaOH and expressed as malic acid content.

Data analysis Statistical analysis was conducted using SAS software (Enterprise Guide 7.1, SAS Institute, USA). The significance of differences between treatments was determined by one-way analysis of variance (ANOVA), and mean differences were analyzed using Duncan's multiple range test (DMRT) at a 5% significance level.

Result and Discussion

Effects of organic amendments on soil physical and chemical properties The physical and chemical properties of soil were assessed in an apple orchard treated with organic amendments (Table 2). The results showed that the NTNA, TNA, LMC, PBC, LTBC1, LTBC2, LTBC3, and MTBC amendments had big effects on the physical and chemical properties of the soil. Soil pH was highest in the LTBC3 treatment (6.36), significantly greater than all other treatments, demonstrating its liming effect. In contrast, the NTNA (6.10) and TNA (5.83) treatments exhibited the lowest pH values, indicating soil acidification in untreated plots. These results aligned with previous findings, which demonstrated that LTBC application effectively increased soil pH levels, contributing to improved soil chemical properties (Yuan and Xu, 2011). The addition of LMC and PBC reduced soil pH compared to LTBC. This reduction in pH corresponded with previous studies, which attributed the decrease to organic acid production during microbial mineralization and nitrification processes (Pattanayak et al., 2001; Yaduvanshi, 2001;

Smiciklas et al., 2002).

Electrical conductivity values exhibited minimal variation across treatments, with LTBC3 and LTBC1 showing slightly higher values in the subsoil compared to other amendments. Deep soil electrical conductivity remained relatively consistent, with no statistically significant differences observed. Organic matter content was highest in the PBC treatment (2.77% in subsoil, 2.56% in deep soil), surpassing all other treatments. Organic matter levels were lowest in the NTNA treatment (1.94% in subsoil, 1.39% in deep soil), highlighting the significant impact of organic amendments on soil organic content. Organic matter content increased with the amount of amendment applied. In the subsoil, organic matter content increased to 2.31% and 2.65% in the LTBC2 and LTBC3 treatments, respectively, compared to 1.39% and 1.48% in the NTNA and TNA treatments (Table 2). Similar findings have been reported in previous studies, which demonstrated that organic amendments improved soil organic carbon and overall fertility by increasing stable organic matter fractions (Steiner et al., 2007; Lehmann et al., 2011b). These results also aligned with observations that BC and compost promoted long-term carbon retention, enhancing soil health and sustainability (Agegnehu et al., 2017).

Table 2. Effects of various organic amendments on the physical and chemical properties of soil collected from mature apple orchard field.

Organic amendments		pH (1:5)	EC (dS m^{-1})	O.M. (%)	P_2O_5 (mg kg^{-1})	Exch. cations ($cmol_c$ kg^{-1})				NH_4^+-N (mg kg^{-1})	NO_3-N (mg kg^{-1})
						K	Na	Mg	Ca		
NTNA	Sub soil	6.10 ABC	0.23 A	1.94 BC	12.2 A	0.56 B	0.065 A	2.96 B	4.73 C	1.12 AB	1.48 B
	Deep soil	5.77 c	0.27 a	1.39 b	12.2 a	0.40 b	0.069 a	3.37 a	4.57 cd	1.05 a	1.18 b
NTNA	Sub soil	5.83 C	0.23 A	1.59 C	14.8 A	0.66 AB	0.073 A	3.37 AB	4.63 C	0.70 B	1.41 B
	Deep soil	5.76 c	0.22 a	1.48 b	12.4 a	0.55 ab	0.075 a	3.27 a	4.40 d	0.75 a	1.90 ab
LMC	Sub soil	5.95 BC	0.23 A	1.85 BC	34.5 A	0.62 AB	0.062 A	3.39 AB	5.33 ABC	0.68 B	1.87 B
	Deep soil	5.77 c	0.24 a	1.74 b	28.4 a	0.41 b	0.071 a	3.41 a	4.88 bcd	0.80 a	2.17 ab
PBC	Sub soil	6.04 ABC	0.23 A	2.77 A	58.9 A	0.67 AB	0.058 A	3.25 AB	6.06 AB	0.63 B	3.36 A
	Deep soil	5.93 bc	0.24 a	2.56 a	27.0 a	0.53 ab	0.074 a	3.28 a	5.57 abc	1.23 a	2.44 ab
LTBC1	Sub soil	6.05 ABC	0.25 A	1.93 BC	73.5 A	0.73 AB	0.055 A	3.31 AB	4.99 BC	0.47 B	1.62 B
	Deep soil	5.93 bc	0.25 a	1.66 b	46.3 a	0.61 ab	0.067 a	3.33 a	4.77 bcd	0.70 a	2.34 ab
LTBC2	Sub soil	6.22 AB	0.20 A	2.51 AB	57.7 A	0.67 AB	0.064 A	3.18 B	6.11 A	1.60 A	1.75 B
	Deep soil	6.15 abc	0.20 a	2.31 a	25.3 a	0.55 ab	0.060 a	3.20 a	5.73 ab	1.33 a	1.78 ab
LTBC3	Sub soil	6.36 A	0.27 A	3.01 A	66.6 A	0.82 A	0.069 A	3.68 A	6.41 A	1.03 AB	1.65 B
	Deep soil	6.47 a	0.21 a	2.65 a	45.1 a	0.65 a	0.066 a	3.56 a	6.13 a	0.84 a	1.41 ab
MTBC	Sub soil	6.31 A	0.23 A	1.87 BC	15.9 A	0.71 AB	0.056 A	3.214 AB	5.40 ABC	1.03 AB	2.33 AB
	Deep soil	6.20 ab	0.22 a	1.66 b	10.9 a	0.64 a	0.059 a	3.20 a	5.073 bcd	0.72 a	2.82 a

Different letters indicate significant difference among treatments at $P < 0.05$ according to Duncan's multiple range test. Uppercase letters (A, B, C) indicate significant differences of sub soils among main treatments, while lowercase letters (a, b, c) indicate significant differences of deep soils among main treatments.

The differences in soil improvement effects between LTBC and MTBC can be attributed to their distinct physicochemical properties, influencing nutrient availability, microbial activity, and carbon stability. LTBC, produced at lower temperatures (300 - 400°C), retains more labile organic compounds, leading to higher microbial

activity and faster nutrient release (Ahmad et al., 2014). In contrast, MTBC, produced at medium temperatures (500 - 600°C), has greater porosity and aromatic carbon content, enhancing soil aggregation, long-term carbon sequestration, and nutrient retention (Novak et al., 2009; Wang et al., 2016). LTBC supports rapid microbial colonization due to its bioavailable carbon, increasing soil respiration and enzymatic activity (Lehmann et al., 2011a), whereas MTBC provides a stable microbial habitat, fostering long-term microbial interactions (Khan et al., 2020). Recent studies have shown that BC from greenhouse crop residues optimally produced at 300 - 400°C enhanced soil pH and nutrient retention (Kim et al., 2022). Similarly, research on BC from agricultural residues has highlighted its potential to improve soil quality and nutrient dynamics, making it a promising amendment for sustainable orchard management (Lee et al., 2023). Additionally, LTBC may slightly acidify soils due to retained organic acids, while MTBC acts as a pH buffer, stabilizing soil conditions over time (Yuan and Xu, 2011). These mechanisms align with the results of this study, suggesting that BC properties play a crucial role in soil enhancement. Future research should focus on long-term effects, particularly on soil carbon sequestration, microbial community dynamics, and the economic feasibility of BC application in orchard systems.

Phosphorus content was highest in the LTBC1 and LTBC3 treatments (73.5 mg kg^{-1} and 66.6 mg kg^{-1}, respectively), significantly exceeding levels observed in the NTNA and MTBC treatments. LTBC3 exhibited the highest levels of exchangeable potassium (0.819 cmolc kg^{-1}), magnesium (3.68 cmolc kg^{-1}), and calcium (6.41 cmolc kg^{-1}), demonstrating its superior ability to enhance soil nutrient availability. Sodium levels were higher in the LMC and PBC treatments compared to other organic supplements. NTNA treatments had significantly lower cation levels compared to all organic amendments. These findings aligned with previous studies, which suggested that the incorporation of BC and other organic amendments enriched nutrient availability and increased soil exchangeable cations (Yuan and Xu, 2011; Mekuria et al., 2014; Yang et al., 2024).

Ammonium nitrogen levels were highest in the LTBC2 treatment (1.60 mg kg^{-1}) compared to the tillage treatment (0.70 mg kg^{-1}). Nitrate nitrogen levels were highest in the PBC treatment (3.36 mg kg^{-1} in subsoil), reflecting differences in nitrogen dynamics among treatments. The total nitrogen content of PBC (0.63 mg kg^{-1}) and LMC (0.68 mg kg^{-1}) was relatively low and had a minor impact on ammonium and nitrate nitrogen content in the soil (Table 2). Previous reports suggested that BC and organic amendments enhanced nitrogen retention and reduced nitrogen leaching due to improved nutrient cycling and microbial activity (Agegnehu et al., 2017; Ibrahim et al., 2020). These findings paralleled the results of this study, where LTBC treatments resulted in improved soil nutrient acquisition. This study highlighted the potential of tailored organic amendment strategies to enhance soil health, improve nutrient retention, and mitigate soil degradation in mature apple orchards.

Physical properties of the soil, like its hardness, bulk density, and depth, were important indicators of its health because they affected root growth, water infiltration, air flow, and the structure as a whole. The average soil hardness varied significantly among treatments. The NTNA treatment recorded the highest soil hardness (2.61 MPa), indicating compacted soil conditions in untreated plots. In contrast, the PBC treatment exhibited the lowest soil hardness (1.42 MPa), demonstrating its effectiveness in reducing compaction (Table 3). This was attributed to

the composition of PBC, which was readily decomposed by soil microorganisms. The increased microbial activity stimulated by PBC likely enhanced soil pore formation, improving soil structure and reducing compaction more effectively than other organic materials (Zhang et al., 2023).

Bulk density followed a similar trend, with NTNA showing the highest value (1.53 g cm^{-3}), while LTBC1 (1.32 g cm^{-3}) and LTBC3 (1.33 g cm^{-3}) displayed notable reductions (Table 3). This indicated the potential of LTBC and organic amendments to enhance soil porosity and reduce compaction. These results aligned with prior studies highlighting LTBC's ability to reduce soil bulk density (Mandal et al., 2021; Kamali et al., 2022). Further, consistent with Li et al. (2024), LTBC enhanced soil porosity and reduced compaction in this study. Treatments with PBC, LTBC, and MTBC achieved soil depths exceeding 70 cm, indicating improved soil structure and permeability. Organic amendments significantly decreased soil hardness and bulk density, enhancing soil conditions essential for plant growth (Rayne and Aula, 2020; Ghadirnezhad et al., 2024). These findings emphasized the role of organic amendments and LTBC in improving root zone depth and water infiltration capacity, supporting sustainable soil management.

Table 3. Effect of different organic amendment treatments on soil penetration resistance, bulk density, and effective soil depth.

Organic amendments	Penetration resistance (MPa)	Bulk density (g cm^{-3})	Effective soil depth (cm)
NTNA	2.61 a	1.53 a	33
TNA	1.76 bc	1.31 b	>70
LMC	2.22 b	1.33 b	45
PBC	1.42 c	1.21 c	>70
LTBC1	1.78 bc	1.32 b	>70
LTBC2	1.70 bc	1.39 b	>70
LTBC3	1.65 bc	1.33 b	>70
MTBC	1.81 bc	1.38 b	>70

Different letters indicate significant differences ($P < 0.05$) among treatments according to Duncan's multiple range test.

Effects of organic amendments on greenhouse gas emission The CO_2 emission rates were monitored over two years following the application of organic amendments, including LMC, PBC, and BC, to the soil in a mature apple orchard. The emission trends (Fig. 1a) show distinct fluctuations in response to the type of amendment and seasonal temperature changes. Carbon dioxide emissions were higher during the warmer months, corresponding to increased microbial activity, and nearly absent during the winter when temperatures were low. This seasonal trend aligns with previous studies reporting that microbial respiration in soils is significantly influenced by temperature, with peak activity observed during the warmer periods of May and June (Pietikäinen et al., 2005; Lee et al., 2009).

PBC exhibited the highest CO_2 emission rates, particularly during the initial application period, indicating their rapid decomposition and high microbial activity. In contrast, BC treatments, especially those produced at medium

temperatures, displayed more stable and significantly lower CO_2 emissions, suggesting their greater resistance to decomposition and potential as a sustainable soil amendment. Both LTBC and LMC showed intermediate emissions, reflecting moderate rates of decomposition and nutrient cycling.

The total CO_2 flux, calculated over the study period, further highlights the differences among the organic amendments (Fig. 1b). Soil treated with PBC exhibited the highest cumulative CO_2 emissions (approximately 60 Mg ha^{-1}), significantly surpassing all other treatments ($p < 0.05$). This can be attributed to the readily decomposable nature of PBC, which are rich in labile organic matter. In contrast, soils treated with BC, regardless of production temperature, demonstrated significantly lower total CO_2 flux compared to PBC. Among the BC treatments, MTBC showed the lowest total emissions, emphasizing its stability and slow mineralization rate. The LMC, along with NTNA plots, exhibited moderate emissions, indicating limited but consistent carbon mineralization over time.

The results underscore the potential of BC, particularly MTBC, as an effective organic amendment for soil carbon management. Its low decomposition rate and stable carbon structure contribute to reduced CO_2 emissions, making it a valuable tool for carbon sequestration. This aligns with previous studies highlighting BC role in mitigating greenhouse gas emissions while enhancing soil fertility (Wang et al., 2018; Gupta et al., 2020; Abhishek et al., 2022). In contrast, while PBC and LMC provide immediate nutrient release and improve microbial activity, their rapid decomposition results in higher greenhouse gas emissions, limiting their long-term effectiveness for carbon storage.

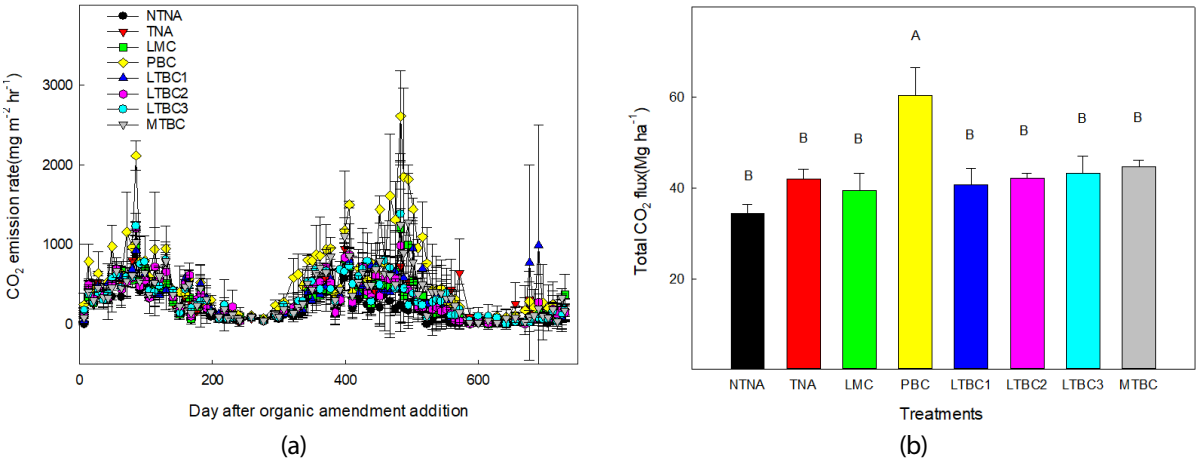


Fig. 1. Effect of organic amendments on CO_2 emission in mature apple orchard soil over two years. The results demonstrate significant differences among NTNA, TNA, LMC, PBC, and various levels of LTBC and MTBC treatments ($P < 0.05$), as determined by a Duncan's multiple range test. (a) CO_2 emission rate (b) CO_2 flux measurements.

There were no significant differences in nitrous oxide emissions across all treatments. While LMC has relatively higher nitrogen content compared to other organic materials, the small amount applied did not lead to significant changes in nitrous oxide emissions (Fig. 2). In PBC and BC made from apple prunes had very low nitrogen content (0.88, 1.00%), and thus did not affect the increase in nitrous oxide emissions. Further research is needed to

investigate nitrous oxide emissions when BC with higher nitrogen content, such as livestock manure BC, is applied.

Our results indicate that MTBC-treated soils exhibited lower CO_2 emissions compared to rapidly decomposing organic amendments such as PBC and LMC, which aligns with previous studies highlighting BC's stability and resistance to microbial mineralization. However, N_2O emissions did not significantly differ among treatments, it is essential to evaluate BC's long-term impact on net greenhouse gas emissions, particularly through its influence on soil nitrogen dynamics and CH_4 fluxes.

These findings suggest that BC, especially MTBC, offers a sustainable solution for reducing CO_2 emissions while enhancing soil health in mature apple orchards. By integrating BC with other organic amendments, orchard management practices can achieve improved soil carbon sequestration, reduced greenhouse gas emissions, and enhanced orchard productivity. Future studies should explore the economic feasibility of large-scale BC applications and their synergistic effects with other organic amendments.

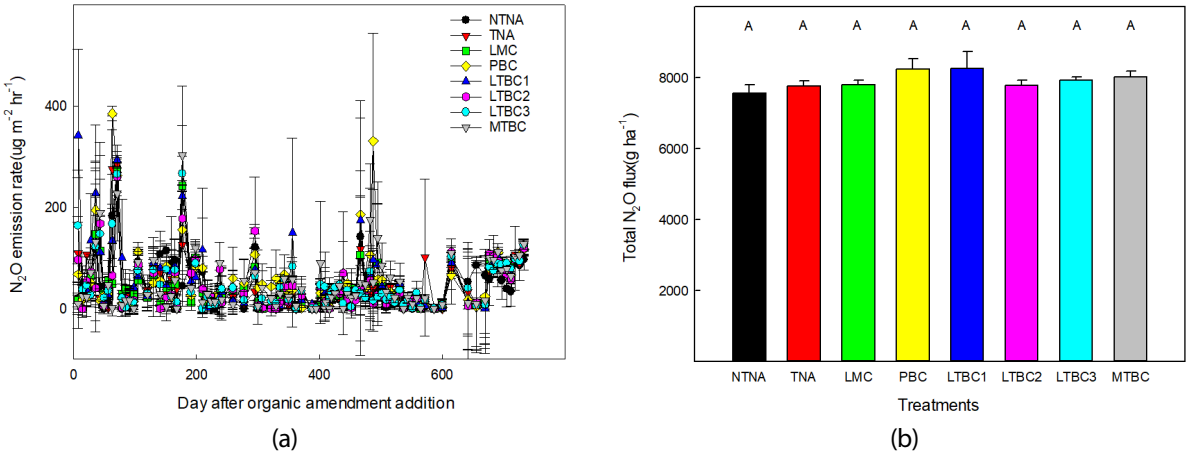


Fig. 2. Effect of various organic amendments on nitrous oxide emission in a mature apple orchard soil over 2-years. All values obtained from each trait were showed significantly different responses under NTNA, TNA, LMC, PBC and various levels of LTBC and MTBC ($P < 0.05$) based on a Duncan's multiple range test. (a) N_2O emission rate (b) N_2O flux measurements.

Effect of organic amendments on vegetative growth parameters Organic amendments play a vital role in enhancing soil quality and tree growth in mature apple orchards. This study evaluated the effects of NTNA, TNA, LMC, PBC, LTBC, and MTBC on tree growth and productivity over a two-year period. Vegetative growth factors, including tree height, canopy width, and stem circumference, were measured in May 2021 and May 2023 following the application of organic materials to the apple orchard soil. As the apple trees were already 7 years old in 2021, direct comparisons were challenging. To address this, growth factor differences over the two years were expressed as ratios, enabling relative comparisons between treatments.

Tree height exhibited varying trends across treatments. NTNA showed a slight reduction (-4.67 cm), reflecting the limited impact of untreated soil on growth. TNA recorded a greater reduction (-6.60 cm), indicating that tillage alone may not significantly improve height without organic amendments. In contrast, LTBC1 and MTBC exhibited

slightly increased tree height (+4.8 cm and +0.6 cm, respectively), suggesting a moderate response to BC. In LTBC2 and LTBC3 treatments showed decreased plant heights (-11.3 cm and -0.5 cm, respectively) (Table 4). While tree height naturally increases with age, farmers often limit it through pruning to manage vigor and simplify orchard operations. Canopy width varied significantly across treatments. NTNA recorded a reduction, while TNA showed substantial increases. LTBC1 and MTBC treatments displayed moderate increases in canopy width (+59.2 cm and +23.8 cm, respectively), indicating improved canopy development with BC application.

Stem circumference remained similar across treatments in 2021 but increased significantly in LTBC2 compared to LTBC3 in 2023 (Table 4). However, these growth differences were challenging to attribute solely to organic amendments, as the trees' age limited the immediate impact of treatments. This aligns with research indicating that while organic amendments, such as livestock manure BC, improve soil quality, excessive application may negatively impact crop growth (Lee et al., 2024b). Similar challenges in short-term evaluations of organic amendments in mature orchards have been noted in previous studies (Steiner et al., 2007; Safaei Khorram et al., 2019; Fornes et al., 2024). These findings suggest that single applications of organic materials may not yield significant growth differences within two years. Future research should focus on applying LTBC and other amendments during the establishment phase of orchards, as younger trees are more responsive to soil enhancements (Ibrahim et al., 2020).

Table 4. Effect of various organic amendment on the growth parameters of matured apple plants.

Organic amendments	May 2021			March 2023			Difference between 2021 and 2023			Ratio
	Height (cm)	Crown width (cm)	Circumference (cm)	Height (cm)	Crown width (cm)	Circumference (cm)	Height (cm)	Crown width (cm)	Circumference (cm)	
NTNA	323.0 ab	218.7 ab	24.4 a	318.3 ab	210.0 c	29.2 ab	-4.67 a	-8.7 a	4.9 ab	0.20 a
NTNA	336.4 a	215.6 ab	29.8 a	329.8 a	265.6 ab	33.7 ab	-6.60 a	50.0 a	3.9 ab	0.13 a
LMC	329.2 a	227.2 ab	28.4 a	323.0 ab	254.0 abc	33.5 ab	-6.20 a	26.8 a	5.1 ab	0.18 a
PBC	324.7 b	214.7 ab	26.2 a	327.3 b	221.3 bc	31.4 ab	2.67 a	6.7 a	5.3 ab	0.20 a
LTBC1	336.0 a	181.4 b	30.7 a	340.8 a	240.6 abc	32.0 ab	4.80 a	59.2 a	1.4 a	0.04 a
LTBC2	330.7 a	213.3 ab	26.3 a	319.3 ab	249.0 abc	37.0 a	-11.33 a	35.7 a	10.7 ab	0.41 a
LTBC3	322.3 ab	226.3 ab	26.7 a	320.0 ab	283.8 a	28.2 b	-2.25 a	57.5 a	1.5 ab	0.06 a
MTBC	335.2 a	236.4 a	28.8 a	335.8 a	260.2 abc	35.8 ab	0.60 a	23.8 a	7.0 b	0.24 a

All values obtained from each trait were significantly different responses under various organic amendments ($P < 0.05$) based on a Duncan's multiple range test.

Influence of organic amendments on photosynthetic activity SPAD measurements, essential for assessing chlorophyll content, directly correlate with photosynthetic efficiency and plant health. Chlorophyll levels influence nutrient uptake, growth, and yield, making SPAD an effective tool for evaluating the impact of organic amendments on plant productivity. In this study, SPAD values showed no significant differences between treatments. However, the highest SPAD values were observed in LMC (48.52), followed by PBC (48.38), suggesting enhanced photosynthetic efficiency and nutrient uptake with these treatments. LTBC3 recorded the

lowest SPAD value (44.13), potentially due to variations in nutrient availability (Fig. 3). These findings align with Montanaro et al. (2017), who reported that single applications of organic materials may not significantly impact mature orchards. Multiple applications or integration with other management practices may be necessary for noticeable changes. Further research is recommended on the use of LTBC in establishing new apple orchards to maximize its benefits.

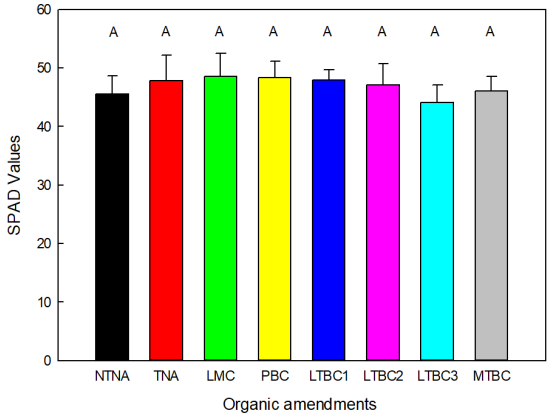


Fig. 3. Effect of various organic amendments on SPAD values in a mature apple orchard. All values obtained from each trait were showed significantly different responses under NTNA, TNA, LMC, PBC and various levels of LTBC and MTBC ($P < 0.05$) based on a Duncan's multiple range test.

Influence of organic amendments in fruit yield and quality The yield and fruit characteristics were assessed across treatments over two years (2021 - 2022). Yield values ranged from 11.1 Mg ha^{-1} in LTBC2 to 16.9 Mg ha^{-1} in MTBC during 2021, and from 40.2 Mg ha^{-1} in PBC to 50.4 Mg ha^{-1} in LTBC1 during 2022 (Table 5). No significant differences in yield were observed among treatments. The reduced fruit yield in the first year was attributed to significant pest issues, particularly anthracnose. In the second year, the yield increased to 50.4 Mg ha^{-1} in LTBC1 and 47.8 Mg ha^{-1} in LTBC3 suggesting the potential of BC to enhance soil fertility and nutrient availability. BC's porous structure and high cation exchange capacity may have contributed to improved nutrient retention and water-holding capacity, thereby supporting higher yields. These findings align with Ray and Bharti (2023), who demonstrated BC effectiveness in improving fruit productivity. Conversely, the relatively lower yields observed in PBC and LMC treatments indicate that their benefits might depend on longer-term soil integration or repeated applications.

Fruit height and width showed minimal variation across treatments. LTBC2 recorded the highest values for fruit height (75.4 mm in 2021) and fruit width (85.5 mm in 2021). However, there were no significant differences in these parameters between treatments over the two years. Fruit hardness ranged from 14.8 N in LMC to 17.3 N in TNA during 2021, and from 14.66 N in LMC to 17.22 N in TNA during 2022 (Table 5). Soluble sugar content varied slightly among treatments, with values ranging from 12.7 °Brix in LTBC1 to 14.3 °Brix in LTBC2 in 2021, and from 13.01 °Brix in LMC to 14.30 °Brix in LTBC3 in 2022. Similarly, fruit acidity showed slight variations

across treatments. MTBC exhibited the highest acidity in both years (0.251% in 2021 and 0.2167% in 2022). LTBC treatments generally showed lower acidity compared to MTBC.

Table 5. Effect of various organic amendments on the yield and quality parameters of apple fruits.

Organic amendments	Yield (Mg ha^{-1})		Fruit height (mm)		Fruit diameter (mm)		Fruit firmness (N)		Soluble solid contents (°brix)		Titratable acidity (%)	
	2021	2022	2021	2022	2021	2022	2021	2022	2021	2022	2021	2022
NTNA	13.8 a	45.5 a	71.5 a	69.2 a	80.2 a	78.8 a	16.7 a	15.77 a	13.1 a	13.70 a	0.200 a	0.1775 b
TNA	11.9 a	46.2 a	72.0 a	72.5 a	82.1 a	81.9 a	17.3 a	17.22 a	14.1 a	13.66 a	0.246 a	0.1695 b
LMC	16.1 a	42.38 a	74.5 a	75.0 a	84.7 a	84.7 a	14.8 a	14.66 a	13.6 a	13.01 a	0.221 a	0.1624 b
PBC	13.0 a	40.2 a	73.5 a	72.4 a	85.4 a	85.6 a	16.6 a	16.48 a	13.8 a	13.80 a	0.211 a	0.1802 b
LTBC1	11.8 a	50.4 a	73.4 a	74.1 a	81.9 a	82.4 a	17.2 a	17.02 a	12.7 a	13.80 a	0.200 a	0.1781 b
LTBC2	11.1 a	46.6 a	75.4 a	74.5 a	85.5 a	84.8 a	16.7 a	16.68 a	14.3 a	13.03 a	0.231 a	0.1675 b
LTBC3	14.6 a	47.8 a	72.1 a	71.2 a	83.5 a	84.0 a	16.0 a	14.82 a	13.8 a	14.30 a	0.222 a	0.1683 b
MTBC	16.9 a	44.3 a	68.2 a	67.9 a	79.6 a	78.3 a	16.0 a	14.72 a	14.2 a	14.23 a	0.251 a	0.2167 a

All values obtained from each trait were significantly different responses under various organic amendments ($P < 0.05$) based on a Duncan's multiple range test.

Fruit hardness and soluble solids content remained stable across treatments, indicating that organic amendments did not negatively affect fruit quality. The higher acidity observed in MTBC aligns with findings by Suthar et al. (2018), which suggest that BC, particularly at higher temperatures, may influence soil pH and subsequently fruit acidity. Similar correlations have been observed in mulberry orchards, where soil organic amendments directly influenced fruit quality (Song et al., 2023). Quality factors such as fruit size, fruit width, hardness, sweetness, and acidity showed no significant differences between treatments. These results suggest that organic amendments had a consistent, though not significant, impact on apple yield and fruit characteristics over the two-year study period (Neilsen et al., 2004). While organic amendments significantly enhance soil fertility, their impact on the yield of mature apple trees is not always evident, particularly in relatively fertile soils within established orchards. This suggests that organic fertilizers may primarily benefit nutrient-depleted soils, whereas their influence on yield improvement in mature orchards may be limited.

Conclusion

This study provides insights into the potential of organic amendments especially BC as an effective soil amendment for enhancing soil properties, carbon management, and tree productivity in mature apple orchards. Our results demonstrate that LTBC significantly improved soil properties, water-holding capacity and nutrient availability, contributing to a more favorable environment for apple tree growth. Additionally, MTBC reduced carbon dioxide emissions, supporting sustainable orchard management. Although, fruit yield and quality parameters remained consistent across treatments, the enhanced soil fertility and reduced greenhouse gas emissions highlight

the environmental benefits of organic amendments. Overall, this study emphasizes the importance of tailored soil amendment strategies to address site-specific soil challenges and achieve long-term sustainability. Future research should focus on combining multiple organic amendments, their application timing, and assessing their economic viability for large-scale adoption. Additionally, exploring the role of BC in newly established orchards could offer insights into maximizing its potential for improving soil health, supporting sustainable apple production, and mitigating the adverse effects of conventional agricultural practices.

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Conflict of Interest

The authors declare no conflicts of interest.

Author Contribution

Rathinapriya P: Writing-original draft, Data curation, **Lee IB & Yi PH:** Validation, Resources, **Jeong ST:** Writing-review & editing, Formal analysis, Supervision.

Data Availability

Not applicable.

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